

Element H

Prototype Test Plan

Ollie L., Ryan M., Monish RJ.

Neuqua Valley High School
PLTW Capstone

Problem Statement

Medical institutions lack a cost-effective, risk-free method for surgical training and other medical processes. Certified cadavers are the best training material for students, but are in great shortage worldwide, and pose ethical and practical issues. Due to said scarcity, organizations such as government agencies allocate a great proportion of their budget towards acquiring these materials: Medicare spends ~\$180,000 per medical resident each year through subsidies to universities, totaling over \$25 billion.¹ Overall, inefficient distribution and lack of product availability hinder surgical trainees worldwide, thus a more accessible solution is required.

I. Product Specifications

To design an effective solution for the problem, specifications must be in place to ensure a quality product that meets consumer needs. The following section denotes the predetermined specifications.

1. Restrictive Force on Fingers

- a. Specification: The solution shall exert a scalable force equal to and opposite to the force applied by the fingers, with a maximum of 8 lbf.
- b. Rationale: The average human can apply 35N (~7.9lbf) of force with one finger². To replicate physical solid touch, the motors must be able to overpower the fingers within a set range of forces.

2. Position and Rotational Accuracy

- a. Specification: The solution shall track the positions of each finger to a sub 5mm accuracy, and the angle of each finger to sub 5° accuracy.
- b. Rationale: Similar solutions such as the Meta Quest 2's hand tracking have an accuracy of 1.1cm and a rotation accuracy of ~10 degrees³. Medical practices require precision, thus the product shall have less margin of error than comparable market products.

¹ "The Costs of Training Internal Medicine Residents in the United States." The American Journal of Medicine, U.S. National Library of Medicine, pubmed.ncbi.nlm.nih.gov/25063649/. Accessed 12 Nov. 2024.

² Riddle, Michael et al. (2020). Evaluation of individual finger forces during activities of daily living in healthy individuals and those with hand arthritis. Journal of Hand Therapy, 33(2), 188-197. [https://www.jhandtherapy.org/article/S0894-1130\(20\)30079-X/abstract](https://www.jhandtherapy.org/article/S0894-1130(20)30079-X/abstract)

³ AbdIkrim, D., Di Luca, M., Aves, P., Maaroufi, M., Yeo, S. H., Miall, R. C., Holland, P., & Galea, J. M. (2024). A methodological framework to assess the accuracy of virtual reality hand-tracking systems: A case study with the Meta Quest 2. Behavior research methods, 56(2), 1052–1063. <https://doi.org/10.3758/s13428-022-02051-8>

3. Element Resistance and Longevity

- a. Specifications: The solution shall stay below 36 °C in use, and withstand prolonged ambient temperatures, different humidities, and water/sweat precipitation.
- b. Rationale: The solution cannot get too hot to prevent uncomfortable heat for the user. Additionally, corrosion and water resistance should be added to ensure excessive dust or sweat does not compromise the functionality of the solution, helping to prolong product lifespan.⁴

4. Weight

- a. Specifications: The solution shall weigh at most 400 grams.
- b. Rationale: Practice surgery sessions can take many hours, and a lightweight solution allows for more natural and unobtrusive usage. Market solutions such as the HaptX Gloves G1 weigh 450g,⁵ hence a comparable yet lighter weight is best.

5. Joint Materials

- a. Specifications: The finger connection points shall be made of a flexible material.
- b. Rationale (Self-Imposed): Flexible materials should be used for the finger mounts and the mounting on the back of the hand to ensure comfortable long-duration sessions.

6. Size Adjustability

- a. Specification: The solution shall be able to fit hand lengths ranging from 5-9 inches.
- b. Rationale: The average male hand length is 7.6 inches and the average female hand size is 6.8 inches⁶. To accommodate this variability, the solution must be able to change length and width to fit most individuals.

7. Battery Life Span

- a. Specification: The solution shall last at least 3 hours of usage time.
- b. Rationale: A standard Meta Quest 3 provides 2-3 hours of battery life on a full charge⁷. To be comparable, the solution shall last the same usage time as the headset.

8. Hot Swappable Battery

- a. Specification: The battery must be replaceable without powering off the solution.

⁴ Gelston, Jon. *When Should You Waterproof Your Electronics Devices? Here's An Answer.*, www.aimprocessing.com/blog/when-should-you-waterproof-your-electronics-devices-heres-an-answer. Accessed 14 Dec. 2024.

⁵ HaptX. (2024). *HaptX Gloves G1*. HaptX. <https://haptx.com/gloves-g1/>

⁶ Sullivan, D. (2020, July 3). Average hand size: Adults, children, and relation to height. *MedicalNewsToday*. Retrieved December 12, 2024, from <https://www.medicalnewstoday.com/articles/average-hand-size>

⁷ Carter, R. (2024, May 6). How to Maximize Meta Quest 3 Battery Life: Top Tips. *XRToday*. <https://www.xrtoday.com/mixed-reality/how-to-maximize-meta-quest-3-battery-life-top-tips/>

- b. Rationale: A hot-swappable battery provides maximum usage by preventing the need to wait for the product to charge. In addition, only the highest-end VR solutions have this ability, so bringing this to a cheaper size creates a serious market advantage.⁸

9. Hidden Cables

- a. Specifications: All wires should be fully encased and routed throughout the solution yet accessible for proper maintenance.
- b. Rationale (Self-Imposed): Exposed wiring causes the ability to snag and damage the product. To ensure the most seamless usage and the longest product life span, the cables shall be hidden yet accessible for maintenance.

10. Connectivity Types

- a. Specifications: The solution must have wired and wireless compatibility with the headset.
- b. Rationale: Wired connections can simplify troubleshooting issues, but a wireless connection is necessary for typical usage. In the market, Bluetooth and Wi-Fi are standard for connecting VR solutions to VR headsets. Products like bHaptics's TactGlove utilize Bluetooth,⁹ while Weart's TouchDiver pro haptic glove utilizes both Wifi and Bluetooth.¹⁰

11. Manufacturing/Repairs and Part Supply

- a. Specifications: The solution must be made of modular replaceable components.
- b. Rationale (Self-Imposed): To keep manufacturing costs down and reparability viable, the product should be composed of 3D-printed parts and replaceable components. If something breaks the cost of repairing the product is minimized. For example, Velcro is a cheap and durable alternative to traditional zip ties or bolts, with a cycle life of around 20,000 uses.¹¹

⁸ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

⁹ bHaptic. (2024). *bHaptics TactGlove DK1 Wireless Haptic Gloves (Pair) - VR Accessory*. KnoxLabs. Retrieved 12 06, 24, from <https://www.knoxlabs.com/products/bhaptics-tactglove-dk1-haptic-gloves?srsId=AfmBOooZhpYmoK6P69NOkd7TPpjTSU4FyB2ZjaAdarD8ZvzBr73iI8YI>

¹⁰ "Weart TouchDIVER Pro Haptic Gloves." *Knoxlabs*, www.knoxlabs.com/products/weart-touchdiver-pro-haptic-gloves?_pos=1&_sid=51b4e4463&_ss=r. Accessed 12 Dec. 2024.

¹¹ Ira, Stephen. "How Long Will It Stick? Understanding the Lifespan of Hook and Loop." *HookandLoop.Com*, 22 Nov. 2023, www.hookandloop.com/blog/how-long-will-it-stick-understanding-the-lifespan-of-hook-and-loop?srsId=AfmBOorZQca6ERDA6PkPlyNbKspVVyYw73iHR_iEGso-OZMVSNLgUUq_. Accessed 14 Dec. 2024.

II. Tested Criteria

Since the prototype solution must be a viable alternative to current market solutions, it should comply with previously set specifications, measured for functionality, accuracy, ergonomics, and other factors. These measured criteria are essential to validate the prototype's core functionality. The following section outlines these specifications, their measurement status, and the rationale behind the decision.

1. Restrictive Force on Fingers

- a. Will be measured
- b. The servos' applied force is the highlight of the product. Without properly applying force to the fingers, the glove cannot properly mimic touch sensation. Measuring force output is a necessity.

2. Position and Rotational Accuracy

- a. Will be measured
- b. The translation of the real-world glove position into the virtual one is a key component that enables the glove to ensure simulation accuracy. As such, measuring this error is crucial.

3. Element Resistance and Longevity

- a. Will not be measured
- b. Typical VR headsets, which this product works in tandem with, are most commonly used indoors and in relatively safe environments. Resistance to these elements is not a crucial test, and not feasible.

4. Weight

- a. Will be measured
- b. Compared to other market solutions, the glove must be lightweight to prevent user fatigue during extensive use. The center of mass is also a great factor in ensuring proper weight distribution.

5. Joint Materials

- a. Will not be Measured
- b. The selection of flexible materials for finger joints was based on prior research. As long as the materials function as expected, direct measurement is unnecessary.

6. Size Adjustability

- a. Will be Measured
- b. Greatly important. The glove must accommodate different hand sizes. Measuring its ability to adjust between 5-9 inches ensures accessibility for most users.

7. Battery Life Span

- a. Will be measured
- b. The glove must last at least 3 hours on a full charge to align with standard VR session durations. Testing the battery duration under typical usage conditions verifies compliance.

8. Battery Hot-Swapability

- a. Will be measured
- b. Swapping batteries without shutting down the system is critical for uninterrupted usage, important for training purposes. Measuring swap time ensures efficiency and usability.

9. Hidden Cables

- a. Will not be measured
- b. The product's appearance is not of the highest concern at this stage in the project, as it is easily changeable, including cable management, thus it is not necessary to measure.

10. Connectivity Types

- a. Will not be measured
- b. The glove will either be wired, wireless, or both, so there is not a proper in-depth test to perform for this specification.

11. Manufacturing/Repairs and Part Supply

- a. Will not be measured
- b. The modular design ensures easy repairability, but this is a qualitative design choice rather than a measurable criterion.

III. Testing Procedures

Test 1: Restrictive Force on Fingers

Primary Tester: Ryan M.

Additional Testers: Ollie L., Monish R. J.

Criteria Tested: #1 - Restrictive Force on Fingers

Incremental Testing Summary:

Conduct tests at varying inputs starting with low voltages and small movements working up to the full 5.5v and complete range of motion movements.

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and additional testing facilities.

Purpose

The purpose of this test is to determine the force exerted by the servo depending on varying input signals.

Pass/Fail Criteria

- Pass: This test is successful if the servos can apply between 7N and 9N of force when at maximum load.
- Fail: This test fails if the servos is not able to apply greater than 7N of force

Materials

- Kitchen Scale (grams).
- Assembled Glove
- 2x Fully Charged Batteries for Glove.

Initial Conditions

- String is tensioned to the correct distance for each finger
- Battery is fully charged.

Safety Concerns

- If the servo is too strong, it could force the fingers to overextend causing potential injury to the tester.

Test Termination

Terminate the test if:

- The servos apply greater than 15N of force.
- Any part of the product begins to smoke.

Data:

The test will produce quantitative and qualitative data. The test will produce a result in weight via the scale in grams, with respect to change in degree of rotation and input voltage. The test will also allow tests to determine if they can overpower the servos.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participant put the glove on and adjust as necessary.
5. Put a scale on a counter and have the participant rest their arm on the same table with their hand resting on the scale.
6. While preventing the hand from moving upwards, Tell the servo to rotate to a different angle, and record the number of grams outputted by the scale.
7. Convert the number of grams to newtons by multiplying by 981 N/g.
8. Record this number in a data sheet and record the angle the servo started at and ended at.
9. Repeat this experiment with varying starting and ending angles, as well as with each finger.

Test 2: Position and Rotational Accuracy

Primary Tester: Monish R.J.

Additional Testers: Ollie L., Ryan M.

Criteria Tested: #2 - Position and Rotation Accuracy

Incremental Testing Summary: Yes

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and Testing Zone

Purpose

The test shall measure the variance in glove movement between the real and digital world.

Pass/Fail Criteria

Position

- Pass: This test succeeds if the digital position of the glove is at most 1 cm off from its real position.
- Fail: This test fails if the digital position of the glove is greater than 1 cm off from its real position.

Rotation

- Pass: This test succeeds if the digital angle of the glove is at most 10 degrees off its real angle.
- Fail: This test fails if the digital angle of the glove is greater than 10 degrees off from its real angle.

Materials

- Assembled Glove (with battery power)
- Oculus Quest Headset (with Test Application running)

Initial Conditions

- Scale in the test application must be on and displaying the digital hand position.
- Headset and controller functioning correctly.

Safety Concerns

- Ensure no excessive strain is placed on the glove's internal components.
- Verify secure attachment of the glove to avoid slippage or misalignment.

Test Termination

Terminate the test if:

- Any part of the product begins to smoke.
- Tracking systems (Oculus Controller) fail to function as expected.

Data

The data is primarily quantitative, capturing incremental changes in position and rotation. Both real-world and digital data will be graphed to analyze discrepancies, ensuring accurate calibration. Specific points in 3D space will be designated for testing and plotted on a coordinate plane to visualize motion. Differences in position and rotation will be graphed and analyzed to identify deviations and estimate the true coefficient difference.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participant put the glove on and adjust as necessary.
5. Begin motion tracking using the Test Application.
6. Have the participant place their hand at a set zero position in the real world.
7. Mark down the digital position of the zero.
8. Have the participant move their hand along predefined real positions with known distances.
 - a. List of predefined positions up to tester, while keeping consistent.
9. Measure the deviation in position and angle between real and digital environments.
 - a. Two separate data sheets for position and rotation,
 - b. Rotation only shall be tested at zero.
10. Record all data on the accompanying datasheet.

Test 3: Weight and Center of Mass

Primary Tester: Ollie L.

Additional Testers: Ryan M., Monish R. J.

Criteria Tested: #4 - Weight and Center of Mass

Incremental Testing Summary: None

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and additional testing facilities.

Purpose

The purpose of this test is to determine the glove's center of mass and weight to ensure comfortable usability in a VR environment.

Pass/Fail Criteria

- Pass: This test is successful if the glove weighs less than 680 grams(~1.5 lbs) and the center of mass is within 2 inches from the user's elbow.
- Fail: This test fails if the glove's total weight exceeds 680 grams or the center of mass is more than 2 inches away from the user's elbow.

Materials

- Digital Scale (grams)
- Balancing Rig
- Assembled Glove
- 2x Fully Charged Batteries for Glove

Initial Conditions

- Assembled Glove with all components strapped and attached
- Battery is equipped

Safety Concerns

- If the glove is not fastened properly, some unsecured components could cause the weight to shift

Test Termination

Terminate the test if:

- Any part of the product begins to smoke
- If the glove causes strain to the user's arm
- If any part of the glove detaches or breaks

Data:

The test will include both quantitative and qualitative data. The test will produce a calculated center of mass in relation to the user's elbow, the weight of the glove in grams, and any observational notes on user comfort. Additional qualitative data can be based on if any part of the glove shifts over time, causing one side of the glove to be heavier than the other.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Setup the balancing rig and determine the center of mass in both the x and y direction.
5. Mark the estimated center of mass location and measure the deviation from the elbow.
6. Weigh the glove itself on a scale.
7. Compare results with expected ergonomic standards.
8. Record measurements in a data sheet and record the weight of the glove.
9. Repeat this experiment with varying finger, wrist, and glove orientations.

Test 4: Size Adjustability

Primary Tester: Ollie L.

Additional Testers: Ryan M., Monish R. J.

Criteria Tested: #6 - Weight and Center of Mass

Incremental Testing Summary: Yes - conduct tests with participants with varying hand sizes and lengths.

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and additional testing facilities.

Purpose

The purpose of this test is to determine whether the glove can accommodate various hand sizes and lengths, ensuring usability for any customer.

Pass/Fail Criteria

- Pass: This test is successful if the glove fits hand lengths ranging from 5-9 inches without significant discomfort or loss of functionality.
- Fail: This test fails if the glove is either too loose or too tight for varying hand sizes or becomes loose over time.

Materials

- Assembled Glove
- Ruler/Measuring tape
- 2x Fully Charged Batteries for Glove

Initial Conditions

- Assembled Glove with all components strapped and attached
- Battery is equipped

Safety Concerns

- Ensure that the glove is not over tightened.
- Verify all attachment points and straps are securely fastened.

Test Termination

Terminate the test if:

- Any part of the product begins to smoke
- If the glove causes the user's hand any pain(poking, compression, restricted circulation)

Data:

The test will include both quantitative and qualitative data. The test will measure a range of hand lengths and sizes, and produce observations on ease of adjustment and proper fit.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participant measure their hand length, width, and thickness.
5. Adjust the glove to fit the hand comfortably.
6. Record participant feedback on comfort, mobility, and adjustment ease in a data sheet.
7. Record success or failure for each hand size.
8. If the glove failed a hand size, record what the problem was and any possible solutions.

Test 5: Battery Life Span

Primary Tester: Ryan M.

Additional Testers: Ollie L., Monish R.J.

Criteria Tested: #7 - Battery Life Span

Incremental Testing Summary: No

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and Testing Zone

Purpose

- The purpose of this test is to determine how long the controller is able to be used for when only limited by the batteries.

Pass/Fail Criteria

- Pass: This test is successful if the battery of the glove lasts longer than 2 hours.
- Fail: This test fails if the battery of the controller dies before 2 hours.

Materials

- Stopwatch (iPhone or similar device with a stopwatch)
- USB-C Charging Cable
- Computer equipped and Bonelabs or Unity.
- Oculus Quest Headset.
- Assembled Glove
- 2x Fully Charged Batteries for Glove.

Initial Conditions

- The string is tensioned to the correct distance for each finger
- Battery of the glove and headset is fully charged.

Safety Concerns

- Due to the long test duration, the servos and microcontroller have the potential to overheat.
- Low battery voltage could cause irregular inputs to the servos which could result in the fingers overextending.

Test Termination

Terminate the test if:

- Any part of the product begins to smoke.
- The servos stopped responding as expected.

Data

The test will result in quantitative data. The stopwatch will record a time that can be averaged to find the average battery life span of the glove.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participant put the glove and vr headset on.
5. Plug the VR headset into the computer.
6. Connect the glove to the headset.
7. Start a stopwatch.
8. Have the participant use the VR Headset and the controller in either Unity or Bone Labs until the controller dies, or stops functioning as expected, when this happens stop the stopwatch.
9. Take off the glove and headset.
10. Record the time on the stopwatch in a datasheet.
11. Charge the batteries up to 100%.
12. Repeat steps 4-11, until enough data has been collected to come to an appropriate average battery life span. (5-10 total trials).

Test 6: Battery Hot-Swapability

Primary Tester: Monish R.J.

Additional Testers: Ollie L., Ryan M.

Criteria Tested: #8 - Battery Hot-Swapability

Incremental Testing Summary: No

Testing Date: February 28th - March 31st

Testing Location: Neuqua Valley High School and Testing Zone

Purpose

To determine whether the battery can be replaced while the glove remains operational without system shutdown or performance degradation.

Pass/Fail Criteria

- Pass: This test succeeds if the glove remains operational throughout the swap, and the swap time is under 10 seconds.
- Fail: This test fails if the glove powers off, resets, or the swap time exceeds 10 seconds.

Materials

- Assembled Glove (with battery power)
- 3x Fully charged batteries (two already plugged in)
- Stopwatch for measuring swap time
- Oculus Quest Headset (with Test Application running)

Initial Conditions

- Test Application is running and measuring glove connection
- Headset and controller functioning correctly.
- The replacement battery is fully charged and ready for swap.

Safety Concerns

- Ensure no excessive strain is placed on the glove's internal components.
- Ensure batteries are handled properly to prevent short circuits.
- Test in a controlled environment to prevent electrical hazards.

Test Termination

Terminate the test if:

- Any part of the product begins to smoke.
- The glove experiences a complete power failure or shuts down unexpectedly.
- The battery fails to provide power after installation.

Data

The data is both qualitative and quantitative, checking to see how much time it takes to swap the battery without system interruption. We also observe system stability and functionality

post-swap and measure in various quantities any delays in responsiveness depending on what parts of the system are lagging.

Stepwise Procedure

1. All materials are on hand.
2. All initial conditions have been verified.
3. All participants are familiar with safety concerns.
4. Have the participant put the glove on and adjust as necessary.
5. Start the stopwatch and remove the current battery
6. Insert the replacement battery as quickly as possible.
7. Stop the stopwatch once the battery is securely connected.
8. Verify the glove has continued its functionality.
9. Check the test application for data on the glove functionality.
10. Ask the participants for any qualitative performance irregularities they noticed.
11. Record all data on the accompanying datasheet.
 - a. If any failure occurs, investigate the cause and document observations.

IV. Data Collection Chart

To ensure the prototype meets design specifications, key performance metrics must be evaluated through quantitative and qualitative data collection, as planned in Section III. The following chart outlines the criteria, required data, measurement type, and accuracy thresholds for those tests.

Criteria/Benchmark	Description of data needed	Quantitative/ Qualitative	Degree of Accuracy
#1 - Restrictive Force on Fingers	Force exerted by servos at various input levels in Newtons. Ability of the glove to resist finger movement without exceeding safe pressure.	Both	$\pm 0.2\text{N}$ of applied force
#2 - Position and Rotation Accuracy	Deviation between real-world finger positions/angles and virtual representation in mm and degrees.	Quantitative	$\pm 5\text{ mm}$ position error, $\pm 3^\circ$ rotation error
#4 - Weight	The total mass of the glove and distribution of weight across the hand and forearm in cm.	Quantitative	$\pm 5\text{ grams}$ total weight, $\pm 1\text{ cm}$ center of mass deviation
#6 - Size Adjustability	Ability to accommodate hand lengths from 5-9 inches. Ease of adjustment and secure fit for different users.	Both	$\pm 0.2\text{ inches}$ fit tolerance
#7 - Battery Life Span	Duration of continuous operation under normal use. Any fluctuations in power delivery affecting glove performance.	Quantitative	$\pm 5\text{ minutes}$ of expected runtime
#8 - Battery Hot-Swapability	Time (in seconds) required to swap the battery without system interruption. Observations on system stability, functionality post-swap, and any delays in responsiveness.	Both	$\pm .3\text{ seconds}$, $\leq .5\text{ seconds}$ of no function (without loss of data)

V. Conclusion

Overall, these tests plan to establish a structured methodology to evaluate the potential VR glove solution on various criteria to ensure quality, functionality, accuracy, and usability. By systematically testing force feedback, motion tracking, weight distribution, size adjustability, battery efficiency, and hot-swap performance, the solution ensures compliance with engineering standards and stakeholder expectations. Collected data will guide refinements to optimize the solution iteratively.